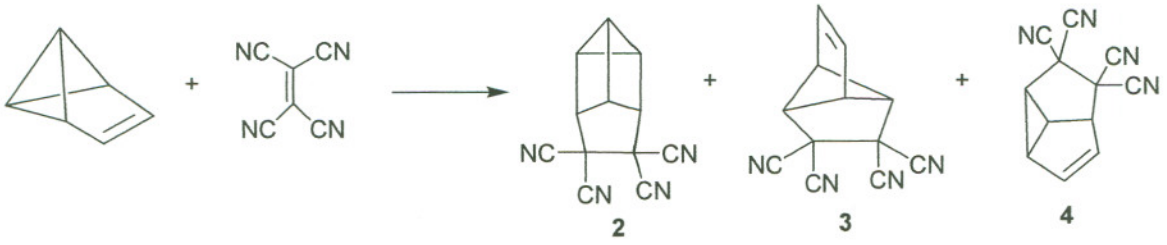
Organic Cume – February 11, 2005

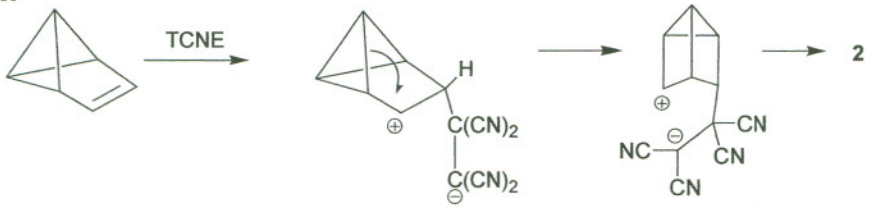
1) The reaction of benzvalene with tetracyanoethylene (TCNE) provides three addition products: **2**, **3**, and **4**.



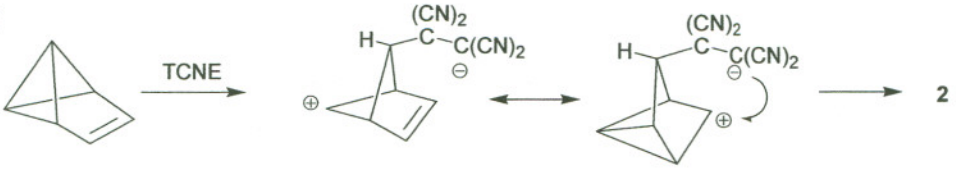
Shown below are mechanisms that might account for the formation of **2**, **3**, and **4**:

For compound **2**:

A:

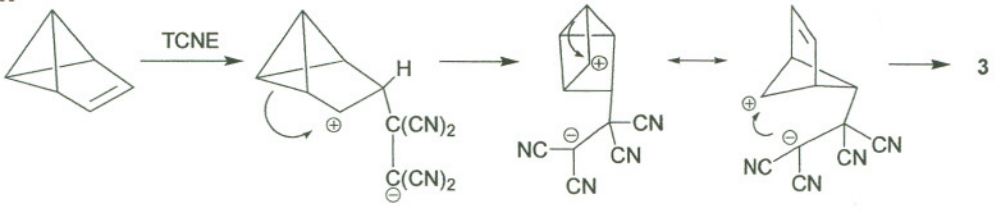


B:

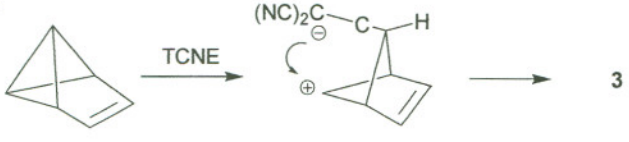


For compound **3**:

A:

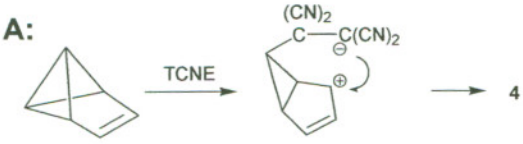


B:

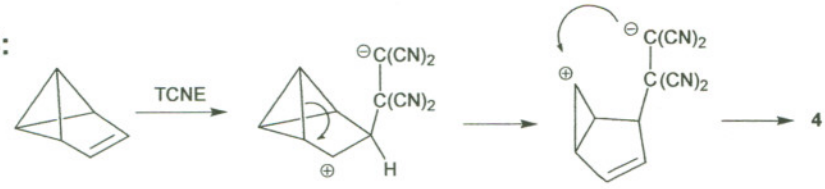


For compound **4**:

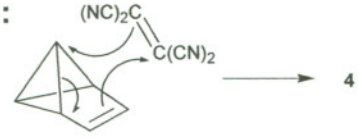
A:



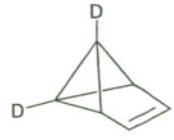
B:



C:



For each of the seven mechanisms above, predict which isotopomers of **2**, **3**, and **4** would be produced from the reaction of TCNE with the 5,6-dideuteriobenzvalene, *i.e.*



When answering this question it is important to keep in mind that the bicyclo[2.1.1.]hex-2-en-yl cation (**5**), and its substituted derivatives, must be represented by three separate resonance structures:

2) Which mechanism(s), if any, **<u>can not</u>** be distinguished using this labeling technique?